

SET - 16

17. Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

AIM: To Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

OUTPUT:

```
D:\Desktop\Query_Processing_Lab>python experiment17.py
Original DataFrame
  School_Code Student_Name  Age
0          S01         Arun   15
1          S02         Priya   16
2          S01         Kiran   14
3          S03         Rahul   17
4          S02         Divya   15
5          S01          Anu   16

Mean, Minimum and Maximum Age for Each School
  School_Code  mean  min  max
S01          15.0   14   16
S02          15.5   15   16
S03          17.0   17   17
```

RESULT: Thus, the Pandas program to split the DataFrame by school code and calculate the mean, minimum, and maximum age for each school was executed successfully.

18. Write a Pandas program to split the following given dataframe into groups based on school code and class.

AIM: To Write a Pandas program to split the following given dataframe into groups based on school code and class.

OUTPUT:

```
Original DataFrame
  School_Code Class Student_Name Age
0         S01   10A         Arun  15
1         S01   10B         Kiran  16
2         S02   10A         Priya  15
3         S02   10B         Divya  16
4         S01   10A          Anu  15
5         S03   10A         Rahul  17

Groups based on School Code and Class:

Group: ('S01', '10A')
  School_Code Class Student_Name Age
0         S01   10A         Arun  15
4         S01   10A          Anu  15

Group: ('S01', '10B')
  School_Code Class Student_Name Age
1         S01   10B         Kiran  16

Group: ('S02', '10A')
  School_Code Class Student_Name Age
2         S02   10A         Priya  15

Group: ('S02', '10B')
  School_Code Class Student_Name Age
3         S02   10B         Divya  16

Group: ('S03', '10A')
  School_Code Class Student_Name Age
5         S03   10A         Rahul  17
```

RESULT: Thus, the Pandas program to split the given DataFrame into groups based on school code and class was executed successfully